

fong-spivak-seven-sketches — formula report

4029 inline formulas (section omitted; 1 did not render), 264 display equations, 63 tables, 503 unrecovered image regions.

Unresolved (1)

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches F03716	317	—	$\left[f ; h ; \iota_{\mathrm{E}}, g \circ k ; \iota_{\mathrm{F}}\right]=(f+g) \boxtimes \text{[U+27C28]}(h+k)$	(not rendered)	$[f \circ h \circ \iota_E, g \circ k \circ \iota_F] = (f + g) \circ (h + k)$, as required.
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Flagged, not acted on — 3 of 72 shown

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches_090054 • C +43	095	■ 0.526	$\mathbf{3} := \operatorname{Free} \left(\begin{array}{c} \stackrel{v_1}{\bullet} \xrightarrow{f_1} \stackrel{v_2}{\bullet} \xrightarrow{f_2} \stackrel{v_3}{\bullet} \end{array} \right)$	$\mathbf{3} := \operatorname{Free} \left(\begin{array}{c} \stackrel{v_1}{\bullet} \xrightarrow{f_1} \stackrel{v_2}{\bullet} \xrightarrow{f_2} \stackrel{v_3}{\bullet} \end{array} \right)$	$\mathbf{3} := \operatorname{Free} \left(\begin{array}{c} \stackrel{v_1}{\bullet} \xrightarrow{f_1} \stackrel{v_2}{\bullet} \xrightarrow{f_2} \stackrel{v_3}{\bullet} \end{array} \right)$
fong-spivak-seven-sketches_090064 • C +31	112	■ 0.147	$\mathbf{Gr} := \begin{array}{c} \text{Arrow} \xrightarrow{\text{source}} \text{Vertex} \\ \text{no equations} \end{array}$	$\mathbf{Gr} := \begin{array}{c} \text{Arrow} \xrightarrow{\text{source}} \text{Vertex} \\ \text{no equations} \end{array}$	$\mathbf{Gr} := \begin{array}{c} \text{Arrow} \xrightarrow{\text{source}} \text{Vertex} \\ \text{no equations} \end{array} \quad \mathbf{DDS} := \begin{array}{c} \text{State} \bullet \\ \text{next} \updownarrow \\ \text{no equations} \end{array}$
fong-spivak-seven-sketches_090164 • C +22	257	■ 0.917	$\stackrel{A}{\bullet} \xrightarrow{\quad} \stackrel{B}{\bullet} \quad \stackrel{C}{\bullet}$	$\stackrel{A}{\bullet} \xrightarrow{\quad} \stackrel{B}{\bullet} \quad \stackrel{C}{\bullet}$	$\stackrel{A}{\bullet} \xrightarrow{\quad} \stackrel{B}{\bullet} \quad \stackrel{C}{\bullet} \subseteq \left \begin{array}{c} \stackrel{A}{\bullet} \xrightarrow{f} \stackrel{B}{\bullet} \xrightarrow{h} \stackrel{C}{\bullet} \xrightarrow{i} \stackrel{D}{\bullet} \end{array} \right $
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured					

The remaining 69 flagged rows are stated as a count.

Each differs from its scan by fewer than 20 components, which is the tail of the distribution rather than its bulk: 37 are component (C), 32 weak (W). By MathPix confidence: 39 at ≥0.9, 12 in 0.6–0.9, 18 below 0.6, 0 with none. Out/465 sampled twenty rows drawn from the ≥0.9 band corpus-wide and found eighteen purely typographic differences, one typographic plus a crop overrun, one borderline and *no* unambiguous content errors — so a high-confidence flag here is evidence of a rendering difference, not of a defect. Every row is in report.ink.json.

Doubted but correct (2)

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches- EQ0133 • S +1	214	■ 0.025	$\begin{aligned} \mu_X &:= \left(X + X \xrightarrow{\mathrm{id}_X} \mathrm{id}_X X \right) \\ \eta_X &:= \left(\emptyset \xrightarrow{!_X} X \xleftarrow{\mathrm{id}_X} X \right) \\ \delta_X &:= \left(X \xrightarrow{\mathrm{id}_X} X \xleftarrow{\mathrm{id}_X} X + X \right) \\ \epsilon_X &:= \left(X \xrightarrow{\mathrm{id}_X} X \xleftarrow{!_X} \emptyset \right) \end{aligned}$	$\mu_X := \left(X + X \xrightarrow{[\mathrm{id}_X, \mathrm{id}_X]} X \xleftrightarrow{id} X \right)$ $\eta_X := \left(\emptyset \xrightarrow{!_X} X \xleftarrow{\mathrm{id}_X} X \right)$ $\delta_X := \left(X \xrightarrow{\mathrm{id}_X} X \xleftarrow{[\mathrm{id}_X, \mathrm{id}_X]} X + X \right)$ $\epsilon_X := \left(X \xrightarrow{\mathrm{id}_X} X \xleftrightarrow{!_X} \emptyset \right)$	$\begin{aligned} \mu_X &:= (X + X \xrightarrow{[\mathrm{id}_X, \mathrm{id}_X]} X \xleftarrow{\mathrm{id}_X} X) \\ \eta_X &:= (\emptyset \xrightarrow{!_X} X \xleftarrow{\mathrm{id}_X} X) \\ \delta_X &:= (X \xrightarrow{\mathrm{id}_X} X \xleftarrow{[\mathrm{id}_X, \mathrm{id}_X]} X + X) \\ \epsilon_X &:= (X \xrightarrow{\mathrm{id}_X} X \xleftarrow{!_X} \emptyset) \end{aligned}$
fong-spivak-seven-sketches- EQ0188 • S +2	279	■ 0.038	$g_{!}(\left(c_3\right)=\{ \}^{\wedge 12}\{ \}_\downarrow^{\wedge 3} \bullet \bullet \bullet \bullet)$	$g!(c_3) = {}^{12} \downarrow {}^3 \bullet {}^4 \bullet$	$g!(c_3) = \begin{matrix} 12 & 3 & 4 \\ \downarrow & \downarrow & \downarrow \\ \bullet & \bullet & \bullet \end{matrix}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					